

Alina Devkota

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Third-year PhD student in Computer Science at West Virginia University with 4 years of industry experience in Machine Learning and Deep Learning (ML/DL)

Research Interests: Multimodal Learning, Out-of-distribution (OOD) Detection, Foundation Models, Federated Learning, Vision-Language Models (VLM), Large Language Models (LLM), Medical Imaging

EDUCATION

Lane Department of Computer Science and Electrical Engineering

West Virginia University, Morgantown, WV, USA

Anticipated Graduation: Dec. 2027

PhD in Computer Science

Department of Electronics and Computer Engineering

Tribhuvan University, Nepal

September 2019

Bachelor's in Computer Engineering

PUBLICATIONS (Full Paper List: <https://scholar.google.com/citations?user=oo43-CUAAAAJ&hl=en&oi=ao>)

- Devkota A., et al. *Federated Foundation Model for GI Endoscopy Images*. **Npj Digital Medicine**, 2026
- Devkota A., et al. *FedVG: Gradient-Guided Aggregation for Enhanced Federated Learning*. **Findings of the Conference on Computer Vision and Pattern Recognition (CVPR)**, 2026
- Devkota A., et al. *AI analysis for ejection fraction estimation from 12-lead ECG*. **Scientific Reports (Nature Portfolio)**, 2025
- Thrasher J., Devkota A., et al. *Multimodal federated learning in healthcare: a review*. **Journal of Healthcare Informatics Research**, 2025
- Thrasher J., Devkota A., et al. *Te-SSL: Time and event-aware self-supervised learning for Alzheimer's disease progression analysis*. **MICCAI**, 2024
- Mu, James, et al. *Towards Federated Learning Across Biobanks: Prototype Software*. **Carnegie Mellon University–NVIDIA Hackathon**, 2026
- Amgain, Sanskar, et al. *Local K-Similarity Constraint for Federated Learning with Label Noise*. **International Symposium on Biomedical Imaging (ISBI)**, 2025

SELECTED PROJECTS

Multimodal Out-of-Distribution (OOD) Detection

Dec. 2025 – Present

- Developing methods to detect out-of-distribution inputs across multiple data modalities, primarily focusing on Vision-Language Models to improve AI robustness and reliability

Jailbreaking Multimodal Unlearning Models

Jan. 2026 – Present

- Investigating jailbreak and adversarial prompting strategies against multimodal unlearning models to evaluate unlearning reliability and identify failure modes under malicious or adaptive attacks

Rationale-Augmented Knowledge Distillation for Small Language Models

Sep. 2025 – Dec. 2025

- Developed a multi-task knowledge distillation framework to train compact language models (e.g., T5-Small, FLAN-T5-Small) to jointly predict task labels and reconstruct LLM-generated rationales

Feature Selection in Manufacturing using Knockoff Framework

Dec. 2024 – Present

- Implemented the knockoff framework for feature selection in manufacturing to select 44 relevant features out of 590 while controlling false discovery rates

Contrastive Learning Framework for Knowledge Distillation

April 2024 - Jan. 2025

- Conducted robustness analysis against adversarial attacks and imbalanced data

AWARDS AND ACADEMIC SERVICES

- **Web and Publicity Chair**, 4th DEMI Workshop, MICCAI 2026
- **Second Place**, Rapid Fire Presentation, Statler College Research Week, West Virginia University, 2026
- **Reviewer**, DEMI 2025 (MICCAI Workshop) and MICCAI 2026
- **Lecturer**, WVU SURE Generative AI Workshop, West Virginia University, 2025
- **Second Place**, Poster Presentation, WVU AI Symposium, West Virginia University, 2025
- **Second Place and Honorable Mention**, Graduate Poster Competition, West Virginia University, 2024
- **Presenter**, Global AI Bootcamp, Nepal Cloud Professionals, 2023
- **Teaching Assistant**, Third Nepal Winter School in AI, NAAMII, 2021
- **Speaker**, Part of Research Training, Fusemachines Nepal, 2021
- **Best Project Award**, Department of Computer Science and Electrical Engineering, Tribhuvan University, 2019
- **Fusemachines AI Fellowship**, Fusemachines Nepal, 2019
- **Winner**, at App-a-thon, Spiralogs International, 2017
- **Academic Excellence Scholarship**, Tribhuvan University, 2015-2019

LEADERSHIP EXPERIENCE

- **Organizer**, LOCUS, Tribhuvan University, 2018 - 2019
- **Volunteer**, Ministry of Education, Nepal, in collaboration with NSDEVIL, 2019
- **Trainer**, CAN Cybersecurity Taskforce, 2018
- **Trainer**, Children in Technology, 2017
- **Department Coordinator**, Kathmandu Valley Leo Club, District 325 A1, 2015 - 2016

EXPERIENCE

West Virginia University, Morgantown, WV

Jan. 2024 – Present

Graduate Research Assistant

- Built and evaluated a diverse set of deep learning and classical models on a 55K-patient ECG cohort to estimate left ventricular ejection fraction in a rural Appalachian population
- Trained a foundation model for GI endoscopy images using federated learning, improving disease classification accuracy by 8.9% while preserving data privacy
- Developed FedVG, a gradient-guided federated aggregation method that adaptively weights clients using global validation gradients, improving generalization across heterogeneous client distribution

coac GmbH, Cologne, Germany

April 2022 – Dec. 2023

Machine Learning Engineer

- Enhanced OCR and object detection in PDFs containing scanned schematic diagrams, improving recognition accuracy and inference speed

- Investigated optimization algorithms to minimize pandemic lockdowns in German counties, and used AI models to replicate a mathematical model to reduce computation time from ~5 hours to ~1 minute with an acceptable tolerance of 5%

NepAI Applied Mathematics and Informatics Institute for Research (NAAMII), Nepal July 2021 – July 2022

Research Assistant, *Part-time*

- Applied UNET for fetal head segmentation to measure head circumference (HC18 dataset)
- Used a classification algorithm to detect cancer in ultrasound images of breasts

Fusemachines, Nepal

Sep. 2019 – April 2021

Machine Learning Engineer

- Led ML pipeline development to reconstruct 3D bones from X-rays (Intelligent surgery project)
- Implemented pseudo-labelling to develop a student status engine (FuseClassroom, an online LMS)
- Integrated Elasticsearch for plagiarism detection, significantly reducing the search space and computation time in large-scale text comparisons

SKILLS

- **Programming:** Python, C, C++, R, JavaScript, MATLAB
- **Machine Learning & Deep Learning:** PyTorch, Scikit-learn, OpenCV, Hugging Face, Transformers, Multimodal Learning, Foundation Models, Federated Learning, VLM, LLM, TensorFlow, LangChain
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib
- **Software & Tools:** Slurm, Git/GitHub, Ray, ClickUp, Jira, Docker, AWS, FastAPI, Django, REST APIs, SQL, MongoDB